Dynamic Hand Gesture Recognition

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***Abstract*— Gestures were most likely utilised by our ancestors to communicate. Armstrong once stated that he believes movements using the hands were the earliest form of complex human communication. The beginning stage of human computer Interactions is a gesture recognition system. Here, we have designed a Dynamic Hand Gesture Recognition (HGR) System using a Neural Network, which can recognize the gesture using the Computer or Laptop's Web Camera and do the corresponding tasks. The model has been divided into mainly three modules. Firstly, Module 1 is about building a CNN model which we use to predict the gestures. Our second module is about predicting the gesture through live video feed. The final module is about assigning a specified task for a particular predicted gesture. For the predicted gesture, we use PyAutoGUI module to control devices like mouse and keyboard. Finally, the system is made to do some desired tasks in response to the gestures and obtains 99.84% of accuracy.**

Keywords— Hand gestures, CNN, System Control, Hand gesture recognition, PyAutoGUI

I. Introduction

Gesture Recognition is an important and active field of Computer Vision. Gestures can be Static or Dynamic (Motion), but direct use of the hand can be a more attractive and natural way of conveying the meaning to the humans and computer. But it also depends on the speed of motion and latency. Gesture recognition has several uses in human-computer interaction, sign language communication, video surveillance, dance/video annotations and forensic identification. This paper is motivated from the problem that is faced by deaf and mute people that is prevailing for a long period. It eliminates the direct use of hands in public places especially during a pandemic situation like this. So, building and designing a system which can detect the gestures and can be widely used for delivering the information or controlling the devices. The introduction to the techniques that we use in this paper are as follows. We will use a database of images that represent the gestures and make a Convolutional Neural Network model, then train the model to classify the images based on the gestures from the database. The next step is to use the trained model to predict the gestures from real time live video feed. Finally, we assign some tasks to perform based on the predicted gestures. There are many applications where hand gestures can be used for interaction such as video games. These hand gestures can also be used by handicapped people to interact with the systems. Classical interaction tools like keyboard, mouse, touchscreen, etc. may limit the way we use the system.

The purpose of this research is to apply transfer learning to develop several CNN models that can detect diverse real-time hand motions. Gestures can interpret the same functionality without physically interacting with the interfacing devices. The problem lies in understanding these gestures, as for different people, the same gesture may look different for performing the same task. This problem may be solved by the use of Deep Learning approaches. Convolution neural networks (CNN's) are proving to be the ultimate tool to process such recognition systems. High computing power is required in order to process gestures with higher accuracy and efficiency. HGR systems are used for interfacing between computers and humans. The objective of this paper is to develop a model for recognition of hand gestures with reasonable accuracy and also to make the system to perform specific tasks based on the predicted gestures. The detailed summary introduces us to the basic knowledge about HGR. It also throws light on the importance of this recognition system and the motivation behind the work. The techniques that were used are also explained along with the problem statement. The final part of the introduction contains the main objective of the work..

II. LITERATURE REVIEW

Hand gestures and gesticulations are a common form of human communication. It is therefore natural for humans to use this form of communication to interact with machines as well. The main objective of this effort is to explore the utility of a neural network-based approach to the recognition of the hand gestures. For instance, touch-less human computer interfaces can improve comfort and safety in vehicles. Computer vision systems are useful tools in designing such interfaces. However, real-world systems for dynamic HGR present numerous open challenges. First, these systems receive continuous streams of unprocessed visual data, where gestures from known classes must be simultaneously detected and classified. Here we have taken a vision Based Model for gesture recognition as the vision based model deals with Video capturing, image processing and pattern recognition. The main objective of this effort is to explore the utility of a neural network-based approach to the recognition of the hand

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gestures. Here we can compare with the other previous works related to HGR. It can also motivate us to understand the various advantages and disadvantages respected with their various techniques. In this literature review, we can discuss various methods for capturing hand gestures, such as using physical gloves [2], or simply using a camera to capture hand motions and map some of the shortcuts for opening applications [1], as well as some other methods that are likely more colour-oriented, such as wearing colored gloves to identify the hand but this document includes four stages for recognising hand motions [3]. The enhanced Adaboost approach is used to identify hands. It is followed by the hand tracking and segmentation operation. At last scale-space features detection is applied to find palm-like and finger-like structures [4]. The coloured video is taken as input and converted into binary [5]. Threshold segmentation is done to the RGB video to convert it to binary video. Expansion and corrosion are done as a part of pre-processing. Contour extraction is done in order to locate the hand. A VGG-Net network is used to classify the contour region as hand or face. Then finally the gesture is recognised [5]. There are several additional ways for recording hand gestures [6], where they employ the Microsoft Kinect to capture the human hand, which offers us with a virtual 3D Skeleton model of the human body, which includes the fingertip. This is how its interpret its gesture, let us see the purpose of the gesture recognition, basically here [6] the gesture trajectory is obtained by the Microsoft Kinect skeleton tracker, it stored in the sequence of the X, Y, Z coordinates to the hand tip of the person, it enabling the algorithm to compare the gestures that is observed independently from the relative point to the person from the sensor [6]. And some other researchers had done with the HGR by using sEMG Signals [7], The gesture acquisition is done by sEMG acquisition equipment to store and filter the electric signals that are generated by skin surface [7], Here they explain why they used the sEMG signal because it is a weak signal and it is easy to interface with the bioelectrical signal. This technology adopts 16-channel high spatio-temporal resolution sampling technology, this equipment can also able to collect the upper limb signals, it accompanies the most of the research tasks that are based on the EMG Signals, And Here they designed a lightweight glove kind of wearable [8]. That glove can capture and send hand motions wirelessly and in real time. They adhered with the IMU sensors on the surface of the fabric in that glove to track the motion of the palm and the motion or the rotations of the hand joints of the fingers [8]. The hand positions and gestures are obtained from the leap motion that is recorded in the coordinate to the centre of the leap motion; they have done the coordinate transformation for embedding the leap motion control based interaction in the existing system. They did the transformation by using the affine transformation method [10]. For the data fusion they used the methods kuke Kalman filter. for combining the data from the leap motion and Myo sensors to create 3D simulation of the Hand motion, Kalman mapping also used to optimise the estimation of hand position and orientation they used the interpolation based resampling for the combining both the sensors and SLR. How they are detecting means by using the LMC-API it allows access to detect the hands or arms and also they have implemented the 21 features [11]. These three fingers have the Raw data returned by IMUs processed by the fusion algorithm and they have done several processing and extracting the hand gestures [8]. Sensors used for HGR include wearable sensors such as data gloves and external sensors such as video cameras. Hand position and movement can be accurately measured using data gloves [9]. But data gloves are very expensive. A detector is used as a switch to activate the classifier if a gesture gets detected. Its job is to distinguish between gesture and no gesture classes by running on a sequence of images. Two models C3D and ResNet are used as a classifier model [11].

Hand gestures and gesticulations are an ordinary form of human ideas. It is so natural for people to use this form of ideas to communicate with accompanying machines as well. This uses the webcam to discover salutes created for one consumer and act elementary operations therefore. From the Existing Research Papers, we raise that Human Computer Interaction has a more important field, One of the Major Fields of calculating apparition in the understanding of human gestures. Hand gestures hopeful a habit of trading news accompanying added crowd in a in essence scope, directing few bots to act sure tasks in surroundings, or interplay accompanying the calculation. In that we have two various fads of recognition like One is the Vision Based model but this depends on the Environment and another is sensor located accompanying wanted environment belongings like lights and unique Environment. In that we have two different modes of recognition like One is the Vision Based model but this depends on the Environment and another is sensor based with needed background things like lights and isolated Environment. These hardware ploys are mainly some sensors or some other somewhat Arduino manoeuvre accompanying some alternative of sensors contained in it. Thus, the interaction between human and calculating becomes less “unaffected” by way of the use of aforementioned devices. Also, these schemes are not inexpensive, aesthetic and smooth to claim. For these reasons, it becomes unsettled too low for people to approach and use bureaucracy with ease. Human Computer Interaction (HCI) electronics are briskly evolving the habit we communicate by accompanying calculating devices and conforming to the uniformly growing demands of new paradigms. One of ultimate beneficial forms concerning this is the integration of Human-to-Human Interaction gestures to further ideas and meaning plans. The big objective of transporting this Systematic Literature Review (SLR) searches out the focal point of the state-of-the-creativity in the circumstances of fantasy-located gesture acknowledgment accompanying distinguishing devote effort to something hand sign acknowledgment (HGR) methods and permissive technologies. After a painstaking orderly excerpt process, 100 studies having to do with the four research questions were selected. While it cannot be of excellent use for skimming computer networks or article a paragraph document, it is beneficial in news performers and while learning documents or files. A plain nod could pause or play the picture or increase the capacity even while situated a great distance away from the calculating screen. One keeps a manuscript through an eBook or a performance even while bearing a luncheon. Existing orders for only numbers and few elementary gestures so we can’t invite to do battle for all, but likely expected change always we have second-hand these gestures to communicate accompanying the calculation. For example, appearance, your touch to increase the book, existent orders for only numbers and few elementary gestures so we can't invite to do battle for all - meaning likely expected change forever. We have used these gestures to communicate while calculating or they were utilising the fittings like Kinect or leap motion that cost about nearly 100 USD that is much costlier in our country and still energetic supplies are skilled but that form more explosion so we have created this plan. We have captured a fantasy Based Model for signal Recognition, Gesture acknowledgment is a numerical understanding of a human motion to a computer, the view-located approach handles the Video Capturing, countenance dispose of and pattern acknowledgment. For this we should use the CNN classifier to decide the shape of the help. In the dream-located approach they have prevented skin colour separation. The aim search out identify active help gestures accompanying asserting the extreme veracity and moment of truth captured to acknowledged gestures from the Beginning they were second hand the sensor located manoeuvres like Use of photoelectric fittings, but it was Complex to use cause the Physical fastening accompanying the model is necessary and more explosion were caused while evolution and they create the protection but it again have a concurrent Problems and Kinect find a resolution but it was more expensive so achieving for all it was more troublesome so we have papered to work the unchanging use in our calculating arrangements accompanying the use of desktop computer or calculating's webcam. We are making use of implementing a plan that identifies Gesture recommendation utilising webcam & performs the Specified Operation. This use may be fashioned to arrest the backdrop while the consumer runs added programs and uses. This is very useful for a hands-free approach. In order to remove the disadvantage of bureaucracy, we can train the model accompanying images that contain few cry secrets. In other words, we can train the model accompanying concepts accompanying the culture. The training dataset concedes the possibility of a merger of concepts accompanying and outside culture. Further tweaks may be organised in the rule to increase the efficiency of the signal acknowledgment process. The rule may be upgraded for better understanding and acknowledgment of the gestures and more recent gestures can be incorporated for more functionalities. The program that controls display for accumulating and inspecting gestures in addition to running the program may be enhanced considerably, such as providing an interactive GUI alternatively utilising terminal commands. By this habit we can build a robust model that acts less computations. So, at last we have papered to work the unchanging service in our calculating arrangements with the use of a desktop computer or calculating's webcam.

III. SYSTEM ANALYSIS

Introduction to our HGR system consist of Convolutional Neural Network to predict the gesture and it was created using the Keras, which is interface of Tensor flow to detecting the gesture and control the operating system according to the assigned task. Existing methods are for only numbers and some basic gestures so we can't do it for all, meaning tends to be changed. We have always used these gestures to interact with the computer. For example, showing your palm to increase the volume, existing methods for only numbers and some basic gestures so we can't do it for all. We have used these gestures to interact with the computer or they were using hardware like Kinect or leap motion which costs around 100 USD which is much costlier in our country and also electrical equipment is there but that makes more noise so we have come up with this idea. We can segment our proposed work in three units/modules – 6 CNN Model Generation. Here, we will use a database of images and make a CNN model, train it to classify the images and any other same kind of image. User Feed Taking: Now, as we have made the CNN model, we will use this model to predict the gesture from real time live video feed. Doing the Task: As the prediction goes on, the system will do the task assigned to each gesture when they are predicted as shown in the Figure 1.

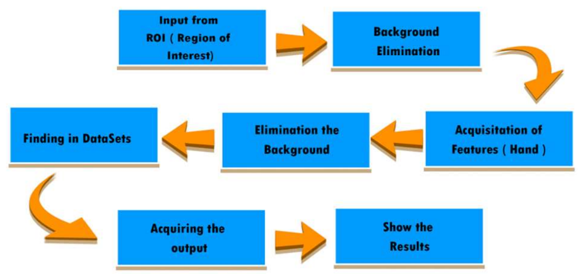

Figure 1 Process for HGR

IV. SYSTEM DESIGN AND IMPLEMANTATION

We will see the systematic design of the work along with the detailed implementation of various modules which are the building blocks of this paper. There are three modules in total and are explained below.

A. Module 1

Module 1 is about building a CNN model which we use to predict the gestures. For that we use Keras (interface for Tensor Flow) to build and train the model using a local database of images. This forms the basis of our work as shown in the Figure 2.

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 98, 118, 32)       320

 max_pooling2d (MaxPooling2D  (None, 49, 59, 32)       0
 )

 flatten (Flatten)           (None, 92512)             0

 dense (Dense)               (None, 128)               11841664

 dropout (Dropout)           (None, 128)               0

 dropout_1 (Dropout)         (None, 128)               0

 dense_1 (Dense)             (None, 64)                8256

 dense_2 (Dense)             (None, 32)                2080

 dropout_2 (Dropout)         (None, 32)                0

 dense_3 (Dense)             (None, 6)                 198

=================================================================
Total params: 11,852,518
Trainable params: 11,852,518
Non-trainable params: 0
_________________________________________________________________
None
```

Figure 2 Keras Model

B. Module 2

Our second module is about predicting the gesture through live video feed as shown in the Figure 3. From that input taken at real time by web camera we predict the gestures by the users by using the trained model. We used OpenCV for the purpose of image and video processing. NumPy is used to make images as arrays for further computations and lastly TensorFlow keras to predict the gesture. The below image shows various steps of processing. It includes converting the image into grayscale followed by the Gaussian blur. Then we find the absolute difference between the background and the current frame which gives the output image.

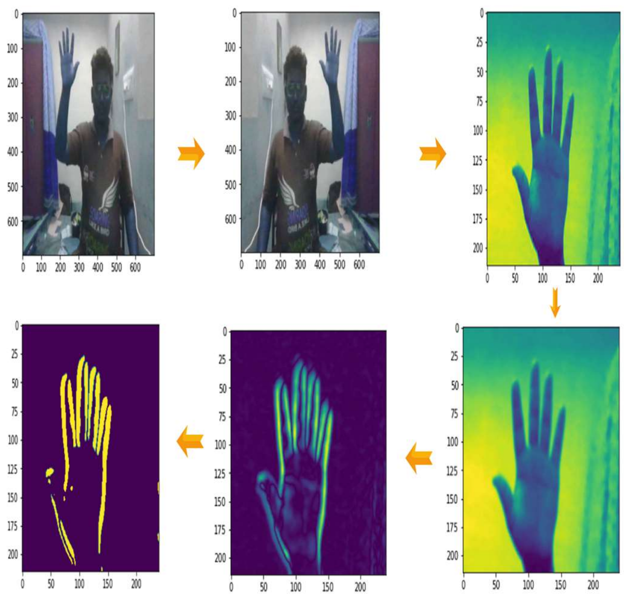

Figure 3 Gesture acquisition in the model

C. Module 3

This is the final module of the work shown in the Figure 4. This module is about assigning a specified task for a particular predicted gesture. We use PyAutoGUI for controlling devices like mouse and keyboard. Finally, the system is made to do some desired tasks in response to the gestures.

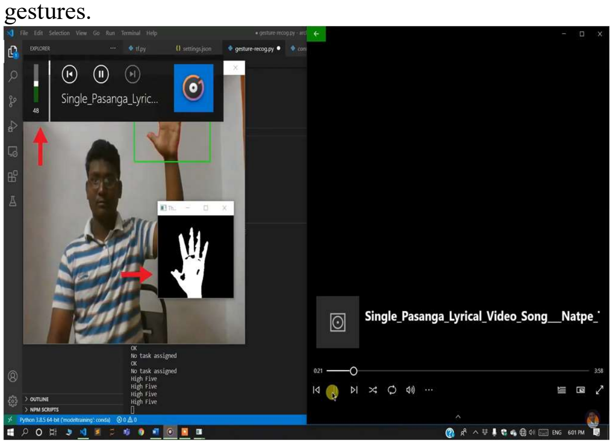

Figure 4 Live Working of the model

V. Performance Analysis

Introduction Here we are going to analyse how our model was performing in the stream of HGR. Our model is giving the accuracy of 99.84% approximately. Here we are providing the chart that shows how our model was performing while training as shown in the Figure 5, Figure 6, Figure 7.

```
Epoch 1/5
197/197 [==============================] - 16s 43ms/step - loss: 24.5892 - a
ccuracy: 0.9456 - val_loss: 0.0000e+00 - val_accuracy: 1.0000
Epoch 2/5
197/197 [==============================] - 8s 40ms/step - loss: 0.2535 - acc
uracy: 0.9940 - val_loss: 0.0000e+00 - val_accuracy: 1.0000
Epoch 3/5
197/197 [==============================] - 8s 40ms/step - loss: 0.1321 - acc
uracy: 0.9968 - val_loss: 0.0000e+00 - val_accuracy: 1.0000
Epoch 4/5
197/197 [==============================] - 8s 40ms/step - loss: 0.0617 - acc
uracy: 0.9976 - val_loss: 0.0000e+00 - val_accuracy: 1.0000
Epoch 5/5
197/197 [==============================] - 8s 39ms/step - loss: 0.0287 - acc
uracy: 0.9984 - val_loss: 0.0000e+00 - val_accuracy: 1.0000
```

Figure 5 Values of epoch and Training and Testing Accuracy

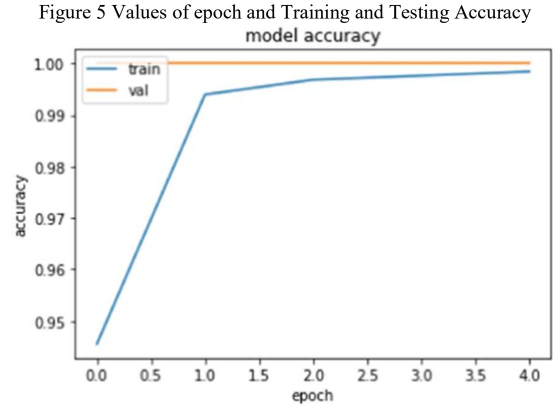

Figure 6 Epoch vs Accuracy

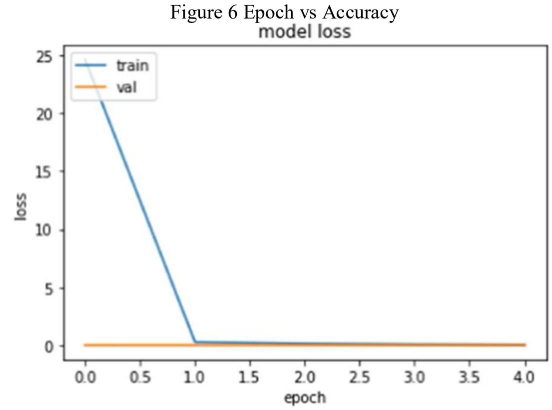

Figure 7 Epoch vs Loss

Here we have included the graph, table and charts, our model has reached the accuracy as 0.9984999895095825 that is approximately 99.84%.

A. Results

In order to eliminate the limitation of the system, we can train the model with images that contain some noise in the background. The training dataset should be a combination of images with and without background to make the model more robust. We can use this dynamic HGR system in industries like for fighter robot, aquatic robot, disease prediction using smart devices and irrigation systems etc., [20-23]. The table here shows some of the work related HGR problems and the comparisons between classification accuracy as shown in the Table 1 and Figure 8.

TABLE 1: COMPARISONS OF CLASSIFICATION ACCURACY

Authors	Methods	Accuracy %
Kim et al., [12]	Using micro-Doppler Signatures with CNN	85.6
Lin [13]	Human HGR using a CNN	95.96
Molchanov [14]	HGR With 3D CNN	77.5
Nagi [15]	Max-pooling CNN for vision-based HGR	96
Zhang [16]	Dynamic HGR based on short-term sampling NN	95.73
Kopuklu [17]	Dynamic HGR based on short-term sampling NN	94.04
Lai [18]	CNN+RNN Depth and Skeleton based Dynamic HGR	85.46
Ahlawat [19]	HGR Using CNN	90.12
Proposed	HGR Using CNN	99.84

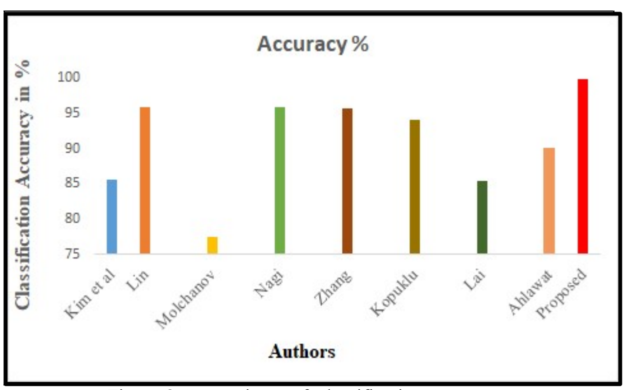

Figure 8 Comparisons of Classification Accuracy.

VI. CONCLUSION

The following part discusses the enhancements that can be made to the system in the future. HGR is inevitable in this modern age of communication. Here we propose some of the limitations that exist and a new way of overcoming those limitations. It is very difficult to build a system for hand gestures with higher efficiency. Because it involves a lot of computations. In many applications, hands only occupy about 10% of the image, whereas the most of it contains background, human face, and human body. So, we need to perform image segmentation and background elimination in real time. This is a constraint of the system. Hand gestures are an intriguing interaction paradigm that may be used in a range of computer applications. Picking a right technology to extract raw hand data is crucial and is also difficult. There are several promising areas of future study in HGR. This survey has provided a comprehensive overview of HGR techniques. To conclude, we can eliminate the existing problem in the system by adding some images to the dataset that contain different backgrounds. Thereby a robust model is built with a reasonable accuracy and involving less computations.

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